

AERODEX

National Airfare Price Index (APIx) & Real-Time Intelligence Platform

Comprehensive System Architecture, Operational Workflow, Feature Matrix & Mathematical Algorithms

Target Institutions:	Ministry of Statistics & Programme Implementation (MoSPI) • DGCA • RBI MPC	Release Version:	v2.4 Enterprise Production
Methodology Scope:	Laspeyres & Fisher Index • COICOP 07.3.3 • DGCA 786-Route Census Basket	Date / Stamp:	September 2026 • Live Nowcast
Core Capabilities:	On-Demand Extraction • Anti-Bot IP Protection • HHI Collusion Watchdog • AI Situation Room	Data Integrity:	Grade A+ (97.7% Integrity Score)

1. Executive Abstract & Institutional Regulatory Mandate

The Macroeconomic Airfare Tracking Problem: The Ministry of Statistics and Programme Implementation (MoSPI) publishes India's headline Consumer Price Index (CPI) with an inherent **30 to 45-day temporal latency**. Under the United Nations standard Classification of Individual Consumption According to Purpose (COICOP 07.3.3: *Passenger transport by air*), price enumerators have historically collected quotes through intermittent monthly manual inquiry. In the modern deregulated Indian aviation market, however, automated airline revenue management engines adjust yield curves multiple times every hour based on flight-load thresholds and advance purchase windows (T+1 to T+45). During festival travel waves (Diwali, Chhath, Durga Puja) or unexpected aviation groundings, domestic airfares surge by **200% to 350%** within hours. By the time central statistical prints are released, the inflationary wave has already passed through the economy unmonitored by the Reserve Bank of India (RBI) Monetary Policy Committee (MPC).

The AeroDex Real-Time Solution: AeroDex is India's first end-to-end, automated, real-time National Airfare Price Index (APIx). The platform combines: (1) **Ethical on-demand web extraction** across major aggregators and airline portals with RFC 9309 compliance and client IP protection; (2) **Census-level spatial weighting** based on official DGCA Form-A annual returns across 786 domestic corridors (136.0 million annual passengers); (3) **Carrier market share volume weighting** tracking active DGCA fleet distributions; (4) **Statutory fare deconstruction** isolating pure airline inventory markup from fuel surcharges and airport taxes; and (5) **Anti-trust collusion monitoring** providing the Competition Commission of India (CCI) and DGCA with mathematical HHI monopoly surveillance.

2. System Architecture & End-to-End Operational Workflow

AeroDex is engineered as a decoupled, multi-tiered enterprise architecture with strict separation of concerns between client presentation, REST API orchestration, anti-bot protected extraction, econometric index processing, and offline auditing.

Architectural Tier	Underlying Technologies	Key Operational Responsibilities & Features
Tier 1: Presentation & UI	Next.js 14 App Router, TypeScript, Vanilla CSS, Lucide Icons	MoSPI Executive Dashboard, Live Pulse Streaming, Dynamic Heatmaps (T+1 to T+45), 1-Click Verification Deeplinks, Historical Shock Replay Studio, Bilingual AI Briefings.
Tier 2: API Gateway	Python Threaded TCPServer, REST Endpoints, SSE Pulse Engine	Routes <code>`/api/v1/search`, `/api/v1/live/pulse`, `/api/v1/macro/timeline`, `/api/v1/compliance/*`</code> . Handles asynchronous thread dispatching and sub-millisecond local cache retrieval.
Tier 3: Egress & Ingestion	scraper.py, proxy_rotator.py, robot_guard.py	On-Demand Extraction (no wasteful background IP hammering), 180s IP-Guard Cache , User-Agent & Sec-CH-UA client hints rotation, RFC 9309 verification, polite token-bucket rate limiting, CAPTCHA/429 interception.
Tier 4: Econometric Engine	index_engine.py, integrity_engine.py	Laspeyres National Basket Aggregation (Base 2024 = 100), Superlative Fisher Indexation, Carrier Market Share Weighting, Lead-Time Surge De-escalation, Tukey 1.5x IQR outlier filter.
Tier 5: Audit Warehouse	SQLite 3 (airfare_index.db), 337,000+ Ingested Quotes	Immutable audit trail of scraped microdata, statutory fare breakdowns, index calculation records, and RFC-4180 CSV export endpoints for ministerial data pipelines.

Tier 6: Machine Learning / AI	Random Forest Regressor, Gemini 3.6 Flash, METAR Radar	9-feature price trajectory regression, severe weather/fog calamity impact models, and autonomous bilingual executive summary memo synthesis in English & Hindi.
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End-to-End Operational Lifecycle (Search to Ministry Export):

1. **User Search Initiation:** Evaluator or citizen queries origin, destination, and future travel date (\$T+\$N\$).
2. **IP-Guard Cache Verification:** Engine checks in-memory cache for \$(Origin, Destination, Date)\$. If scraped within the last 180s, fresh quotes are served in < 15ms, entirely shielding the host IP from redundant requests.
3. **Ethical Gating & Header Shuffling:** If network fetch is needed, RobotGuard verifies robots.txt. ProxyManager rotates desktop User-Agents, Sec-CH-UA client hints, and Google referer context.
4. **Live HTTP SSR Scrape:** Scraper executes lightweight fast SSR parsing against Google Flights, EaseMyTrip, or Cleartrip, extracting carrier names, schedules, stops, and gross tariffs.
5. **Statutory Fare Decomposition:** Bundled prices are deconstructed into Base Fare (~70%), Fuel Surcharge YQ (~18%), Airport Fees (~7%), and GST (5%).
6. **Non-Parametric Cleaning:** Quotes pass the 5-stage cleaning pipeline (2.5x Median Rule, IQR bounds, sold-out removal).
7. **Econometric Indexing & Logging:** Carrier-weighted fare (\$P_{r,t}\$), route index (\$I_r\$), and national basket index (\$I_{APIx}\$) are calculated and committed to SQLite.
8. **Live Telemetry Broadcast:** Results update the real-time pulse stream (/api/v1/live/pulse), refreshing the Next.js UI, macro charts, and AI situation room briefs.

3. Comprehensive Platform Feature Catalog (12 Core Modules)

AeroDex incorporates 12 modular components specifically engineered to address every facet of Problem Statement SIH26056:

Module 1: Real-Time Flight Search & Live Fare Verifier

Queries any domestic corridor on future travel dates. Executes real-time live extraction with provenance badges (■ Live Web Scraped), carrier logos, full schedules, and 1-click deeplinks pre-filled to official airline portals for instant auditor cross-checking.

Module 2: Statutory Fare Deconstruction Protocol

Decomposes bundled consumer airfares into Base Tariff, Fuel Surcharge (YQ), Airport User Fees (UDF/PSF), and statutory GST. Crucial for DGCA and MoSPI to determine whether ticket price surges reflect airline inventory markup or external jet fuel (ATF) excise shocks.

Module 3: National Airfare Price Index (APIx) Laspeyres Engine

Computes official Laspeyres composite price indices calibrated to Base Year 2024 = 100.0. Incorporates 786 domestic routes weighted by 136M annual DGCA passenger journeys, providing daily, weekly, and monthly aggregate indices with zero reporting lag.

Module 4: MoSPI CPI Sub-Group 07.3.3 Sensitivity & Macro Impact

Translates live airfare volatility directly into basis points impact on India's All-India Combined CPI basket (weight: 0.077%). Provides the RBI MPC with real-time early warning of transportation inflation passthrough.

Module 5: 11-Source Governance Coverage Matrix (6 OTAs + 5 Airlines)

Maintains complete compliance transparency across MakeMyTrip, EaseMyTrip, Cleartrip, Yatra, Ixigo, Goibibo, IndiGo, Air India, Air India Express, Akasa Air, and SpiceJet, distinguishing between Active Live Scraping, Ethically Gated routes, and 1-Click Verification Deeplinks.

Module 6: Sector Dynamic Pricing Heatmap (T+1 to T+45)

Interactive advance purchase price matrix mapping airfare escalation curves across Metro-to-Metro, Tier-2, and Island/High-Altitude corridors from next-day departure (T+1) to 45-day advance booking.

Module 7: Pan-India State-Level Airfare Inflation Nowcast

Spatial econometric heatmap across 34 Indian States & Union Territories, contrasting official historical MoSPI transport inflation against AeroDex's live high-frequency airline nowcast to detect regional cost discrepancies.

Module 8: CCI & DGCA Route Monopoly & Anti-Trust Watchdog

Computes the Herfindahl-Hirschman Index (HHI) for every domestic corridor in real time. Flags anti-competitive market concentration (HHI > 2,500) and algorithmic tacit collusion where competing carriers synchronize peak surge pricing.

Module 9: Real-Time Econometric Data Integrity Score (Grade A+)

Autonomous 4-dimensional statistical audit (Fidelity, Cleanliness, Multiplicity, Freshness) scoring data trustworthiness between 0 and 100. Includes a tamper-evident SHA-256 cryptographic governance seal certified under IMF and MoSPI CPI manuals.

Module 10: Predictive Calendar & Aviation METAR Calamity Radar

Integrates live airport METAR weather radar with a predictive calendar mapping gazetted festivals (Diwali, Chhath, Durga Puja) and historical weather disruptions to provide econometric surge risk projections.

Module 11: Historical Aviation Crisis Simulation Studio (`shock\_replay.py`)

Allows policymakers to stress-test historical aviation shocks: May 2023 Go First Grounding (demonstrating Laspeyres substitution bias vs. Superlative Fisher), April 2019 Jet Airways collapse, and June 2022 Post-Ukraine ATF jet fuel spikes.

Module 12: Autonomous Bilingual AI Situation Room

Powered by Gemini 3.6 Flash with a 0ms deterministic local econometric fallback. Automatically synthesizes executive situation room briefs with one-click toggling between formal English and pure Parliamentary Hindi (■■■■■■ ■■■■■■).

4. Mathematical Formulations & Algorithmic Foundations

AeroDex implements rigorous econometric methodologies aligned with the International Monetary Fund (IMF) Consumer Price Index Manual (2020) and the National Statistical Office (NSO) standards:

4.1 Carrier Market-Share Weighted Route Tariff ($P_{r,t}$)

$$P_{r,t} = \sum_{c \in C_r} (\omega_c \cdot p_{r,c,t}) / \sum_{c \in C_r} \omega_c$$

Where C_r is the set of airlines operating on route r ; $p_{r,c,t}$ is the median economy fare quoted by carrier c at time t ; and ω_c is the carrier's domestic capacity weight from DGCA monthly census returns: **IndiGo (6E): 61.2%, Air India (AI): 14.3%, AI Express (IX): 6.4%, Akasa (QP): 4.8%, SpiceJet (SG): 4.2%.**

4.2 National Airfare Price Index (APIx) Laspeyres Aggregation

$$I_{APIx, t} = \sum_{r=1}^R W_r \cdot [P_{r,t} / P_{r,0}] \times 100$$

Where $R = 786$ domestic scheduled routes; $P_{r,0}$ is the calibrated base year fare (2024 = 100.0); and W_r is the route expenditure weight calibrated from DGCA Form-A traffic returns (136.0M passengers): $W_r = (Q_{r,0} \cdot P_{r,0}) / \sum (Q_{k,0} \cdot P_{k,0})$, such that $\sum W_r = 1.0$ (100.0%).

4.3 Constant-Quality Quality Adjustment (Lead-Time Surge De-escalation)

$$P_{r,t}^{cq} = P_{r,t}(T+N) / M(T+N)$$

To prevent acute short-term booking scarcity (e.g. T+1 emergency booking) from biasing the structural national inflation trend, tariffs are normalized by calibrated lead-time surge multipliers: **T+1: 1.52x, T+7: 1.32x, T+15: 1.16x, T+30: 1.05x, and T+45: 0.98x.**

4.4 Superlative Fisher Ideal Price Index (Substitution-Bias Corrected)

$$I_{Fisher, t} = \sqrt{ I_{Laspeyres, t} \times I_{Paasche, t} }$$

During structural shocks (e.g. airline groundings), fixed-basket Laspeyres overstates inflation by assuming consumers still attempt to purchase tickets on grounded planes (+11.2 to +14.8 pts). The geometric Fisher mean dynamically re-weights observed consumer substitution toward surviving carriers.

4.5 MoSPI Headline CPI Sensitivity & Basis Points Passthrough

$$\Delta CPI_{bps} = [(I_{APIx, t} - 100.0) / 100.0] \times w_{Airfare} \times 10,000$$

Where $w_{Airfare}$ = 0.00077 (0.077% weight in MoSPI All-India Combined CPI basket). A +30% national airfare surge produces a **+2.31 basis points** increase in headline national inflation.

4.6 Herfindahl-Hirschman Index (HHI) & Anti-Trust Collusion Risk Score

$$HHI_r = \sum_{c=1}^N (s_{r,c})^2, \quad Risk_{col} = 0.40 \cdot HHI + 0.35 \cdot CV_{fare} + 0.25 \cdot DepSync$$

Where $s_{r,c}$ is carrier c's flight share percentage on route r. Corridors with HHI > 2,500 are flagged as highly concentrated. The Collusion Risk Score combines HHI, fare price clustering (low coefficient of variation), and departure schedule synchronicity.

4.7 Real-Time Econometric Data Integrity Score (0 - 100 Grade A+)

$$Integrity = 0.35 \cdot S_{Fidelity} + 0.25 \cdot S_{Cleanliness} + 0.20 \cdot S_{Multiplicity} + 0.20 \cdot S_{Freshness}$$

Fidelity checks statutory identity $|Total - (Base + YQ + UDF + GST)| \leq 1.0$ INR; Cleanliness checks Tukey 1.5x IQR adherence; Multiplicity verifies cross-corroboration; Freshness penalizes quotes older than active reporting cycles (< 180 min). Certified by SHA-256 cryptographic seal.

4.8 5-Stage Non-Parametric Data Cleaning & 2.5x Corridor Median Rule

$$Threshold_{Upper} = \max(\text{₹}28,000, 2.5 \times Median(P_r)) \quad [\text{₹}35,000 \text{ for Leh/Port Blair}]$$

Eliminates luxury business class seats and charter quotes without biasing skewed exponential fare distributions. Drops null/negative prices, cancels out sold-out cards, and restricts valid domestic coach tariffs to ₹1,500 - ₹95,000.

4.9 Random Forest Price Trajectory Regressor (9 Features)

$$P = f_{RF}(LeadTime_{days}, Distance_{km}, Month, DayOfWeek, CarrierShare, HubOrigin, HubDest, WeekendFlag, HistoricMedian)$$

Trained on 14,500+ empirical SQLite observations. Estimates expected market equilibrium fares across any domestic route. Combined with calibrated shock multipliers for festival periods (M_{event} : 1.20x to 1.90x) and weather disruptions ($M_{calamity}$: 1.50x to 1.80x).

5. Ethical Scraping, Anti-Bot & Client IP Protection Protocol

A critical innovation in AeroDex is its dual-layer anti-blocking architecture that guarantees continuous live extraction while strictly safeguarding the host machine's residential/client IP address from bans or CAPTCHAs.

Anti-Blocking Safeguard	Technical Mechanism	Impact on IP & Infrastructure
1. On-Demand Scraping Exclusivity	Continuous background auto-hammering is strictly disabled. The scraper executes only when an active user searches .	Eliminates 300+ automated bot queries/hour on the developer IP, keeping traffic indistinguishable from human browsing.
2. 180s IP-Guard Cache	In-memory memory cache with a 3-minute TTL keyed on (Origin, Dest, Date).	Filters, airline tabs, or repeat searches are served in under 15ms with zero outbound HTTP calls .
3. User-Agent & Client-Hints Rotation	Rotates 6 curated modern desktop browser profiles keeping User-Agent, sec-ch-ua, and platform synchronized.	Defeats passive TLS fingerprinting and header anomaly detectors deployed on major airline OTAs.
4. Polite Jitter Pacing (RobotGuard)	Token-bucket rate limiter enforcing minimum 3.0s crawl delay with randomized 0.3s-0.7s human jitter.	Avoids machine-speed request burst signatures that trigger Google Flights and Akamai edge rate limiters.
5. Challenge Interception & Auto-Cooldown	Detects HTTP 429, Cloudflare Turnstile, and Google unusual traffic signatures.	Halts execution immediately, quarantines egress node for 180s, and falls back to SQLite warehouse rather than burning the IP.
6. Enterprise Proxy Pool Support	Supports comma-separated rotating proxy lists via the PROXY_POOL or HTTP_PROXY environment variables.	Enables seamless enterprise scaling across commercial rotating proxy networks for national cloud deployments.

The 11-Source Governance Coverage Matrix:

Portal / Carrier	Category	Scraping Protocol	Robots.txt & Status	Governance Mode
Google Flights	Aggregator	HTTP SSR + Playwright	Allow: /travel/flights	Active Live Scrape
EaseMyTrip	Domestic OTA	Playwright DOM Extractor	Allow: /FlightList/Index	Active Live Scrape
Cleartrip	Domestic OTA	Playwright DOM Extractor	Allow: /flights/results	Active Live Scrape
SpiceJet (SG)	Direct Carrier	Playwright Booking DOM	Allow: /search	Active Live Scrape
Ixigo	Domestic OTA	1-Click Deeplink Verification	Disallow: /flights/search	RFC 9309 Gated
MakeMyTrip	Domestic OTA	1-Click Deeplink Verification	Protected (Edge Firewall)	Deeplink Fallback
Yatra	Domestic OTA	1-Click Deeplink Verification	Protected (Edge Firewall)	Deeplink Fallback
Goibibo	Domestic OTA	1-Click Deeplink Verification	Protected (PerimeterX)	Deeplink Fallback
IndiGo (6E)	Direct Carrier	Aggregator Ingested + Deeplink	Protected (Direct Edge)	Aggregator Ingested
Air India (AI)	Direct Carrier	Aggregator Ingested + Deeplink	Protected (Direct Edge)	Aggregator Ingested
Akasa Air (QP)	Direct Carrier	Aggregator Ingested + Deeplink	Protected (Gateway 504)	Aggregator Ingested

6. Institutional Impact & National Implementation Roadmap

Actionable Value Proposition for Ministry Stakeholders:

1. **MoSPI Data Innovation & Ingestion Division (DIID):** Modernizes consumer price index collection with automated, census-weighted microdata ingestion pipelines, delivering 100% genuine market quotes with cryptographic audit certainty and zero survey latency.
2. **Reserve Bank of India (RBI) Monetary Policy Committee:** Provides high-frequency forward nowcasts of transport inflation, empowering the central bank to separate transitory crude oil excise spikes from core domestic passenger demand pressures.
3. **Directorate General of Civil Aviation (DGCA) Tariff Monitoring Unit:** Equips regulators with live corridor dynamic pricing heatmaps, statutory fare decomposition (Base vs. Fuel Surcharge), and advance warning of predatory fare surges.
4. **Competition Commission of India (CCI):** Automates Herfindahl-Hirschman Index (HHI) monopoly detection and schedule synchronicity tracking to police algorithmic tacit collusion in concentrated airline corridors.

Document Certified by: Team AeroDex (Smart India Hackathon 2026)	Target Problem: SIH26056 (MoSPI)	Verification URL: http://localhost:8000/api/v1/live/pulse
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